
Integrable Systems — Exercise Sheet 10

The coordinate Bethe ansatz

Variant: Questions only

Conventions, as in Lecture 10: $\hbar = 1$; the Heisenberg ring of N sites is $H = J \sum_n (\mathbf{S}_n \cdot \mathbf{S}_{n+1} - \frac{1}{4}) = \frac{J}{2} \sum_n (P_{n,n+1} - 1)$, $J \neq 0$ real, with $P_{n,n+1}$ the exchange of the spins at n and $n+1$ and site $N+1$ the same as site 1. M is the number of down spins, $|x_1, \dots, x_M\rangle$ has them at $x_1 < \dots < x_M$, and the shift moving every spin one site on multiplies a state of momentum K by e^{-iK} . Rapidities: $e^{ik} = (\lambda + i/2)/(\lambda - i/2)$; $S(\lambda) = (\lambda - i)/(\lambda + i)$; $\theta_n(x) = 2 \arctan(2x/n)$.

Exercise 1 (the model on a few sites). (i) For two spins $\frac{1}{2}$, show that $\mathbf{S} \cdot \mathbf{S}' - \frac{1}{4} = \frac{1}{2}(P - 1)$ by applying both sides to $|\uparrow\uparrow\rangle$ and $|\uparrow\downarrow\rangle$. (ii) Lecture 10 counts $\binom{N}{M} - \binom{N}{M-1}$ multiplets of spin $\frac{N}{2} - M$, $0 \leq M \leq \frac{N}{2}$. List the counts for $N = 4$ and $N = 6$, and check that they account for all 2^N states. (iii) Show that each bond term $\frac{J}{2}(P_{n,n+1} - 1)$ has eigenvalues 0 and $-J$. Deduce that for $J < 0$ the state $|\uparrow\uparrow\rangle$ with every spin up is a ground state, and for $J > 0$ a highest level.

Exercise 2 (one magnon, and the rapidity). (i) Show that $H|x\rangle = \frac{J}{2}(|x-1\rangle + |x+1\rangle - 2|x\rangle)$ for one down spin at x , and that $|k\rangle = \sum_x e^{ikx}|x\rangle$ with $e^{ikN} = 1$ has $E(k) = J(\cos k - 1)$. (ii) Show that $e^{ik} = (\lambda + i/2)/(\lambda - i/2)$ exactly when $\lambda = \frac{1}{2} \cot \frac{k}{2}$, for $0 < k < 2\pi$, and that then $E = -2J/(4\lambda^2 + 1)$. (iii) Show that $S^+|k\rangle = 0$ for $k \neq 0$ and not for $k = 0$, and identify $|k = 0\rangle$.

Exercise 3 (two magnons: the S-matrix). (i) Lecture 10 shows that Bethe's two-magnon ansatz $\psi(x_1, x_2) = A_{12}z_1^{x_1}z_2^{x_2} + A_{21}z_2^{x_1}z_1^{x_2}$, $z_j = e^{ik_j}$, solves the equations exactly when the contact condition $\psi(x, x) + \psi(x+1, x+1) - 2\psi(x, x+1) = 0$ holds for all x . For finite rapidities, show that it holds exactly when $(\lambda_1 - \lambda_2 + i)A_{21} = (\lambda_1 - \lambda_2 - i)A_{12}$, that is, $A_{21}/A_{12} = S(\lambda_1 - \lambda_2)$ where $\lambda_1 - \lambda_2 \neq -i$ and $A_{12} \neq 0$, and $A_{12} = 0$ at the pole $\lambda_1 - \lambda_2 = -i$. (ii) Show that $|S(\lambda)| = 1$ for real λ , that $S(\lambda)S(-\lambda) = 1$, and that $S(0) = -1$. What does $S(0) = -1$ say about two magnons with equal rapidities? (iii) Show that at total momentum $K = k_1 + k_2 = \pi$ the energy $E(k_1) + E(k_2)$ is $-2J$ however K is split between the two. Why does this make the absence of new momenta after a collision a non-trivial fact?

Exercise 4 (four sites, two magnons, completely). Take $N = 4$ and $M = 2$. (i) Solve the logarithmic Bethe equations $N\theta_1(\lambda_j) - \sum_{l \neq j} \theta_2(\lambda_j - \lambda_l) = 2\pi I_j$ for real roots with the quantum numbers $\{I_1, I_2\} = \{\frac{1}{2}, -\frac{1}{2}\}$: show that the roots are $\lambda = \pm \frac{1}{2\sqrt{3}}$, and find E and K . (ii) Write $\sum_n \mathbf{S}_n \cdot \mathbf{S}_{n+1} = \frac{1}{2}(\mathbf{S}^2 - \mathbf{S}_A^2 - \mathbf{S}_B^2)$ with $\mathbf{S}_A = \mathbf{S}_1 + \mathbf{S}_3$ and $\mathbf{S}_B = \mathbf{S}_2 + \mathbf{S}_4$, and use it to find the energies of the two spin-0 multiplets, the highest-weight states with $M = 2$. Which one is the Bethe state of (i)? What does Lecture 10 say about the other?

Exercise 5 (the logarithmic form). (i) Show that $(x + in/2)/(x - in/2) = -e^{-i\theta_n(x)}$ for real x , and take logarithms of the Bethe equations $\left(\frac{\lambda_j + i/2}{\lambda_j - i/2}\right)^N = \prod_{l \neq j} \frac{\lambda_j - \lambda_l + i}{\lambda_j - \lambda_l - i}$ for real roots to get $N\theta_1(\lambda_j) - \sum_{l \neq j} \theta_2(\lambda_j - \lambda_l) = 2\pi I_j$, and show that $I_j \in \mathbb{Z} + \frac{N-M+1}{2}$. (ii) Using $k = \pi - \theta_1(\lambda)$, derive $K = \pi M - \frac{2\pi}{N} \sum_j I_j$ modulo 2π . (iii) Show that for one magnon the equation is $kN = 2\pi n$, and compare it with the Bohr–Sommerfeld rule of Lecture 6 for a particle going once round a ring of circumference N .

Exercise 6 (every state of six sites with two magnons). (Computational: Lecture 10's table and figure of the $N = 6$, $M = 2$ check.) (i) At each total momentum $K = 2\pi m/6$, find all

the regular solutions of the Bethe equations for $M = 2$ from the polynomial of Lecture 10's tutorial, $-(w + 1)z^7 + 2wz^6 + 2z^2 - (w + 1)z = 0$ with $w = e^{iK}$ and $z = e^{ik_1}$: discard $z = 0$ and $z = 1$, and keep the roots whose two rapidities are finite and distinct. Add the singular pair ($E = -J$, $K = \pi$). (ii) Separately, diagonalise H on the 15 states with two down spins, keep the eigenvectors killed by S^+ , and sort them by momentum. Compare the two lists. (iii) Find the quantum numbers I_j of the real solutions. (iv) Plot the nine levels in the plane of K/π and E/J : the exact diagonalisation as rings, the Bethe solutions as dots marked by type (real, string, singular). This is Lecture 10's figure of the check.

Exercise 7 ★. *Bethe vectors are highest weight, for two magnons.* Lecture 10 cites the theorem that a non-zero Bethe vector satisfies $S^+\psi = 0$. Prove it for $M = 2$. Let $\psi = \sum_{1 \leq x_1 < x_2 \leq N} \psi(x_1, x_2) |x_1, x_2\rangle$ with Bethe's ansatz, $z_1, z_2 \neq 1$, $A_{21}/A_{12} = S(\lambda_1 - \lambda_2)$, and the periodicity conditions $A_{12} = A_{21}z_1^N$, $A_{21} = A_{12}z_2^N$ of Lecture 10. (i) Show that $S^+\psi = \sum_y \left(\sum_{x < y} \psi(x, y) + \sum_{x > y} \psi(y, x) \right) |y\rangle$. (ii) Sum the geometric series, and show that the coefficient of every $|y\rangle$ vanishes.